CHAPTER 9 · CONCEPT 2

Saturation, Concentration and Solubility

A solvent can only hold so much solute — until it's saturated — and heating usually lets it hold more.

How much can dissolve?

Keep adding salt to water and at some point it stops dissolving and settles at the bottom.

Unsaturated	Saturated
Can still dissolve more solute (at that temperature)	Cannot dissolve any more — extra solute settles (at that temperature)

- **Concentration** = amount of solute in a fixed quantity of solution/solvent.
- **Dilute** = little solute; **concentrated** = much solute. (These are relative terms.)
- **Solubility** = the maximum amount of a solute that dissolves in a fixed quantity of solvent (at a given temperature).

Effect of temperature

For **most solids, solubility increases as temperature rises** (hot water dissolves more baking soda than cold). So a saturated solution can dissolve more if you heat it — it becomes unsaturated.

> **Tip for the tea problem:** if sugar won't dissolve, heat the tea or add more water — both let more sugar dissolve.

Where students lose marks

> **Trap 1 — Saturation is "at a given temperature".** A saturated solution can take more solute if you heat it.

> **Trap 2 — Dilute/concentrated are about amount of solute, not volume alone.** Compare solute per unit solvent.

> **Trap 3 — Solids: solubility usually rises with temperature.** (Gases are the opposite — see next concept.)

Model answers — write them like this

Example A. Which is more concentrated: 2 spoons of salt in 100 mL or 4 spoons in 50 mL?

Step 1: Compare solute per mL: 2/100 = 0.02; 4/50 = 0.08.

Step 2: 0.08 > 0.02.

∴ 4 spoons in 50 mL is more concentrated.

Example B. Sugar stops dissolving in your tea. How can you dissolve it?

Step 1: Solubility of solids increases with temperature.

Step 2: Heat the tea (or add more water).

∴ Heating (or adding more water) dissolves the extra sugar.

Ready to practise? → Open the worksheet